

# S P E C T R A V I D E O

## U S E R S G R O U P --- W g t n

Volume 3, Number 3 -- November 1986

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The next meeting is on Monday, the 17th of November  
1986. (ie as usual the 3rd Monday of the month.)  
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NOTE - VENUE FOR NOVEMBER MEETING WILL NOT BE THE CLINICAL SCHOOL  
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Ross -- Public Domain CP/M

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### Club Sponsor

Action Computers -- P.O Box 7442, Wellington South.

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NOTE - VENUE FOR NOVEMBER MEETING WILL NOT BE THE CLINICAL SCHOOL  
SEE YOU AT NEWLANDS INTERMEDIATE  
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EXPLORING BASIC - PART 3  
BY L. A. Dunning

Most of the early home computers had what was called a "Memory Mapped" Screen - that is, one that used on-line RAM to store the display shown. a good example is the TRS-80 Model I which held the screen memory between 15360 and 16376. This enabled smart programmers to use all sorts of tricks/routines to manipulate the display. On the Spectravideo however, the display is not memory-mapped, rather a custom built system is used. This system consists of a TMS 9918/28 or 29 Video Display Processor, 16k of Video Memory, interface ports and software 'Drivers'.

The Video Display Processor comes in three variations so to work with different TV systems, however the function remains the same for each type. The ports referred to are 4 of the 256 input/output lines which the Z80 Central Processing Unit can handle. The 'drivers' are software routines buried deep in the BASIC language, however I've uncovered enough of these to produce this article.

The 9900 series of Processors were designed to free RAM to the user, and simplify complex video displays. There are four modes of display available, of which three are supported on the Spectravideo. These Modes are;

GRAPHICS I : This is the unsupported Mode. The screen is divided into a grid of 32 x 24 positions, each position being an 8 x 8 pixel square. Up to 256 different patterns can be used and put into any position. There are however only 32 different foreground-background colour combinations and these are allocated to 32 groups of 8 patterns.

GRAPHICS II : This is SCREEN 1. The display is divided in the same manner as GRAPHICS I, however there are a total of 768 different patterns, one for each different position. Also, rather than 32 colour combinations each pixel row of each position may have a different foreground and background color.

TEXT : This is SCREEN 0. The display is divided into a 40 by 24 pixel grid, enabling more words on the display than in either GRAPHICS mode. Each position is 6 x 8 pixels and there are only 256 different patterns available. Only two colors are allowed.

MULTICOLOR : This is SCREEN 2. The display is divided into a 64 x 48 grid of chunky blocks (each 4 x 4 pixels). Any of the 15 colors may be used for each block. This mode at first appears to be the most limited, however it can be manipulated with great ease.

SPRITES : These can be used in all modes except TEXT Mode. The Sprite 'Drivers' can, however be used in this mode, to the users advantage.

To maintain the image the 9900 requires either 4k or 16k of extra (dedicated) memory, and uses 8 internal 'registers'. Let us examine each of these registers. Table 1 gives a schematic representation of the registers, and a key to each function:

TABLE 1

| REG<br>--- | BITS (msb to lsb) |    |    |    |    |   |    |    |
|------------|-------------------|----|----|----|----|---|----|----|
|            | 7                 | 6  | 5  | 4  | 3  | 2 | 1  | 0  |
| 0          | 0                 | 0  | 0  | 0  | 0  | 0 | M3 | EV |
| 1          | K                 | BL | IE | M1 | M2 | 0 | SZ | MG |
| 2          | 0                 | 0  | 0  | 0  | N  | N | N  | N  |
| 3          | C                 | C  | C  | C  | C  | C | C  | C  |
| 4          | 0                 | 0  | 0  | 0  | 0  | P | P  | P  |
| 5          | 0                 | S  | S  | S  | S  | S | S  | S  |
| 6          | 0                 | 0  | 0  | 0  | 0  | G | G  | G  |
| 7          | F                 | F  | F  | F  | B  | B | B  | B  |
| VDP        | FL                | 5S | C  | X  | X  | X | X  | X  |

Registers 0 to 7 are 'WRITE ONLY' registers. Their current value cannot be read but the default values set by BASIC may be altered by 'writing' to them.(:Ed.)

M3, M2, and M1 Determine the Mode of Display, as shown below (BASIC amends these values according to the current 'SCREEN'):

TABLE 2

| DISPLAY MODE | M3 | M2 | M1 |
|--------------|----|----|----|
| GRAPHICS I   | 0  | 0  | 0  |
| GRAPHICS II  | 1  | 0  | 0  |
| MULTICOLOR   | 0  | 1  | 0  |
| TEXT         | 0  | 0  | 1  |

EV enables an external Video input. EV = 1 makes the external display active, EV = 0 makes the display inactive.(BASIC sets a default value of 1.)

K determines the maximum memory available to the Video Display Processor. K = 1 means 16k, K = 0 means 4k.(BASIC sets the default value of 1.)

BL is an enable for the Display. BL = 0 will display only the Border color, BL = 1 enables the display. (BASIC leaves the display enabled.)

IE enables the Video Interrupt ( used by Spectravideo to update the variable 'TIME', and for the 'ON INTERVAL' command). IE = 1 enables the Interrupt; 0 disables it. (BASIC enables it).

SZ determines the Sprite Size. SZ = 0 is 8 x 8; 1 is 32 x 32. (BASIC defaults to 8 x 8 Sprite size, and adjust this value according to the SCREEN command option).

MG determines Sprite Magnitude. MG = 0 is normal; 1 is Double Size. (BASIC defaults to normal size, but amends this value in accordance with the SCREEN command option).

NNNN defines the base address for the 'NAME TABLE' ie multiply this value by 400H (=1024) to find its starting location in Video memory.

CCCCCCC defines the starting address for the 'COLOR TABLE' (Multiply by 40H = 64 to find the Video Memory address)

PPP defines the starting address for the 'PATTERN GENERATOR TABLE' (Multiply by 800H = 2048 to find the Video Memory address)

SSSSSSS defines the starting address for the 'SPRITE ATTRIBUTE TABLE' (Multiply by 80H = 128 to find the Video Memory address)

GGG defines the starting address for the 'SPRITE PATTERN GENERATOR TABLE' (Multiply by 800H = 2048 to find the Video Memory address)

FFFF defines the foreground colour while in text mode.

BBBB defines the background colour in Text mode, or the Border colour in any other mode.

The VDP register is a 'Read Only' register.

FL is set to 1 at the end of each (Television) raster scan, and reset to 0 when a number of conditions are met.

C is a flag which is set to 1 if two Sprites coincide.

5S is a flag that is set to 1 whenever there are 5 or more sprites on the same (pixel) line - XXXXX gives the number of the fifth sprite.

How is the Video Display Processor connected to the Spectravideo? The Z80 uses Input/Output 'Ports' 128, 129, 130/131 and 132 (= 80H, 81H, 82/83H and 84H) to interface with the VDP. There are four operations possible using these Ports; you can write a Byte into one of the 8 registers, write a Byte to Video Display Memory, read a Byte from the status (VDP) register or read from Video Display Memory. Table 3 outlines the procedure (in BASIC):

TABLE 3

| FUNCTION          | BASIC TO ACHIEVE THIS FUNCTION              |
|-------------------|---------------------------------------------|
| Write to Register | OUT 129, D : OUT 129, RO                    |
| Write to VRAM     | OUT 129, LB : OUT 129, HO : OUT 128, D      |
| Read VDP          | D = INP( 130 ) (... or INP ( 131 ) ...)     |
| Read from VRAM    | OUT 129, LB : OUT 129, HA : D = INP ( 132 ) |

D = Byte Written/Read  
LB = VDP address Least significant Byte

RO = Register number ORed with 128  
HO = VDP Address Most Significant  
Byte ORed with 64  
HA = VDP Address Most Significant  
Byte ANDed with 63

It's possible to manipulate the display using only these four routines. When writing or reading from Video RAM you can use an auto incrementing function so that after each Read or Write, the VRAM address is increased by one, requiring only the last operation (OUT 128, D or D = INP (132) ) to be performed once the address is set. Unfortunately reading the Status register is not practical in BASIC due to its slowness.

Following articles will look at each Mode in more detail and explain how the SpectraVideo supports each.

+ + + + +  
C L A S S I F I E D S

CENTRONICS Printer Interface - Cost - \$110 - (Negotiable)

Ring Chris McIntyre - 759393 (Home)  
737777 (Work - leave a message)

If moving up to a printer now or in the future you will need a SV802 Interface. These are difficult to obtain and at around \$250-00 Brand New are not cheap. This one is still in the sealed bag and in its box !

FOR SALE

>>>>>>> SV601 EXPANDER  
WITH  
SINGLE DISK DRIVE INSTALLED

> > > > > OFFERS ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?

PHONE Allan McPherson  
789-417

THE FOLLOWING PROGRAMS IN ORIGINAL PACKAGING ARE FOR SALE, AT GIVE AWAY PRICES.

NO. PRICE. NAME OF GAME

1. \$9.00 KUNG FU MASTER
2. \$6.00 BIORHYTHM
3. \$9.00 TELEBUNNIE
4. \$6.00 ARTIST
5. \$9.00 NINJA
6. \$5.00 ARMoured ASSAULT
7. \$5.00 SPECTRON
8. \$8.00 SPECTRA CHECKBOOK
9. \$5.00 INTRODUCTION TO BASIC
10. \$5.00 SPECTRA HOME ECONOMIST
11. \$5.00 FINANCIAL CALCULATOR
12. \$8.00 SPECTRA FILE CABINET
13. \$8.00 LEUCOCYTE (FROM W.A.)
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20. \$5.00 CAR ACE
21. \$9.00 SASA
22. \$5.00 SPECTRA FONT EDITOR
23. \$15.00 SECTOR ALPHA (CARTRIDGE)
24. \$15.00 MUSIC MENTOR (CARTRIDGE)
25. \$15.00 FLIPPER SLIPPER (CARTRIDGE)

IF YOU INCLUDE \$1.00 FOR POSTAGE, I'LL SEND IT CERTIFIED

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H ZUYDAM  
241 HAMILTON ROAD  
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N.S.W. 2165.

## A note to would-be contributors

Don't let Anyone or Anything put you off. The newsletter is made from your contributions, if we don't receive any, we can only produce a poor magazine.

Presently we are lacking in BASIC programs and in hints to beginners, and from beginners. We try to cater to all club members, but lately our we have been receiving mainly specialty items, which are of little benefit to many. Among the many things we would like to see arriving in the mail (addressed to The Editors) are:

- Programs - Basic, Pascal, Assembly, You Name It.
- Programming hints, and techniques.
- Utilities - any type, as long as they are self explanatory or come with instructions.
- News - anything you think other club members might like to know.
- Problems - we will find someone to answer them if we can't.

Everything we receive will be considered.

## Public Domain CP/M

Anyone wanting anything from the library of CP/M PD software, please write to Ross Fletcher at the club address, specifying either which disc, or which programs you would like. The listings in previous issues are printed from double sided disks therefore as most drives are single sided, a whole disc from the list now occupies up to 2 full discs. Most of the programs on the discs are 'Squeezed', this means they have been run through a function in New Sweep (NSW, Nsweep etc. but not the earlier Sweep) so as to take up as little room as possible. They all have a 'Q' as the 2nd letter in their extension, and can be easily unsqueezed by using New Sweep to tag them and then the 'Q' option to unsqueeze them, usually onto another disc. Please send discs, rather than asking us to purchase them for you, and please send them formatted. Please also include, either a stamped return mailer, or add 60c to your payment to cover postage. We will return your disk in your packaging.

## A N A P O L O G Y :

I had intended to have a review ready of Elliot Young's program - SVI FILE DEVELOPMENT SYSTEM. HOWEVER I have been unable to have more than a cursory look before going to press. I can say, however, what I have seen has impressed me. Have a look at the summary of functions which Elliot offers - - - - > Pg 8

# SVI FILE DEVELOPMENT SYSTEM

## SUMMARY OF PROGRAM FUNCTIONS

Version 1.00 13 October 1986 Copyright(C) E. Young

### PROGRAM PURPOSE:

: To enable users of an SVI disk system (with 80 column card and 21K free memory in basic) further access to their files, and to perform many disk tasks to save time and repetition. The program is easy to use and user-friendly.

### SOME OF THE FUNCTIONS:

- Shows a directory display, including:
  - Disk heading
  - Wasted space
  - Full & Used average size
  - Free space (tracks, % and K)
  - Disk status
  - Status of each file
  - Every program grouped according to file type (ASCII, etc)
  - Size of each program to x.xx tracks, or x.x Kilobytes.
- Allows a heading for each disk
- Load, Delete, Protect, Unprotect and Rename any file at the press of one key.
- Shows a statistical breakdown of the files on a disk.
- Views any file on the monitor.
- Full multi-program operations, including delete, protect and unprotect.
- A comprehensive disk map, showing percentage of each track used.
- Easy writing and editing of disk IPL.
- A complete disk test.
- One key changing of disk status.
- Formats a disk.
- Writes onto disk sectors to stop unnecessary disk spin.
- Searches for a filename or part of a filename on many disks.
- Allows the examining of any disk sector.
- Adjusts a disk for PAL-80 (anti-wobble 80 column screen).
- Prints out the directory display to any printer.

Comes complete with 5 pages of documentation, including information on disk errors and double-siding your disks, all for only \$5.00

This program has taken many months to write, and as I am not a professional programmer, is available at this low price. It is a must for all disk users.

### Contact:

Elliott Young  
194 Breaker Bay Road  
Wellington 3  
PH: WN 887-327

## Basic Demonstrations

Free tracks: 6 - Disk status: R/O Protected

### BASIC PROGRAMS

|            |        |           |        |            |        |
|------------|--------|-----------|--------|------------|--------|
| demo       | : 1.8K | demo3     | : 2.3K | demo2      | : 1.3K |
| demo1      | : 0.5K | pipe drw: | 0.2K   | F mac:     | 1.0K   |
| colourspr: | 1.0K   | patterns  | : 1.0K | lo-reslin: | 0.5K   |
| sine       | : 0.5K | SV-2 put: | 0.5K   | house      | : 3.8K |
| tune       | : 0.5K | SV put:   | 0.7K   | bird       | : 0.7K |
| tabletcad: | 5.3K   |           |        |            |        |

### MACHINE CODE PROGRAMS

|    |        |    |        |    |        |
|----|--------|----|--------|----|--------|
| D1 | : 8.7K | D2 | : 8.7K | D3 | : 8.7K |
| D4 | : 8.7K |    |        |    |        |

### ASCII DATA

SP dat: 2.8K

### SCREEN PICTURES

cat pic: 14.5K

## GRAPHICS & TABLET

Free tracks: 1 - Disk status: Unprotected

### BASIC PROGRAMS

|            |        |            |        |            |        |
|------------|--------|------------|--------|------------|--------|
| pad        | : 0.29 | amazr      | : 2.47 | temp       | : 1.00 |
| screendmp: | 0.58   | lgraphbig: | 0.11   | draw bak:  | 1.64   |
| longpr     | : 0.11 | 3print     | : 0.11 | drawer     | : 1.64 |
| lgraph     | : 0.11 | move scr:  | 0.23   | directory: | 3.11   |

### SCREEN PICTURES

|     |           |            |      |            |      |
|-----|-----------|------------|------|------------|------|
| Mel | pic: 3.35 | house pic: | 3.35 | NEWS1 PIC: | 3.35 |
| MUM | PIC: 3.35 |            |      |            |      |

A Plea . . . . .

## RADIO ACCESS

### BROADCASTS FOR

### CHRISTMAS

As its getting near THAT time of the year again, we want to make our last 'Broadcast' for 1986 appropriate to the season.

If anyone has a lack of Inspiration how about some

Christmas Music

Christmas Scenes (Drawing program screen saves or whatever)

Holiday-time Programs

for the December Radio Access Broadcast. Please contact Garry, or send something along to our November meeting.

## DISK UTILITY -- DU-V83

or

things that Spectravideo hasn't told you!!

by Graeme Dick

There is a marvellous public domain programme available, called DU-V83, which enables you to look at any sector on a disk and to modify it if desired. This includes the ability to read and write to the system tracks on the disk. It will read both CP/M and Basic disks (the disk format is the same), and provides the easiest way to permanently alter function key assignments or anything else that you wish to change.

If you are uncertain about how CP/M stores information on disks and accesses that information, a bit of delving with DU will soon clarify that. A little explanation about this will make the purpose of this article clearer. The standard single-sided Spectravideo disk format has 40 tracks (numbered 0-39), with 34 sectors per track, each sector being 128 bytes long. A track can therefore store 4.25k of information ( $34 \times 128/8$ ). Track 1 differs from the rest by being single density, and DU shows that every second sector on this track is empty (null-filled).

How do you get the total disk storage space of 155k, when calculation shows that there is 157.25k on the 37 non-system tracks? The answer to this is that the first 16 sectors of track 3 are set aside for the directory. Each directory entry uses 32 bytes, giving a maximum of 64 directory entries. Taking this away leaves 155.125k.

With disk Basic, the minimum space taken up by a file is 1 track. CP/M works differently, using what is called 'Allocation Groups'. For single-sided Spectravideo disks an allocation group is 8 consecutive sectors, and the minimum space for a file is one Group (1k =  $8 \times 128$  bytes). If you transfer a 1k file from single-sided format to double-sided format, you will find that it now takes 2k of disk space (the minimum file size) - this is because an allocation group is 16 sectors for the double-sided format.

How does CP/M know where to find files on disk? Let's have a look at a portion of the Directory track, for a single-sided disk:

This is what you will see - FIRST LINE

```
00 0046494C454E414D4554595 00 00000 23 .FILENAME.TYP...#
```

```
!! !!                !! !!      !!
!! !!^--hex filename/type^ !! !!      ^file name^
!! !!                !! !!      in ASCII
!! !!                extent^^      !!
!! !!                !!
!! !!                file size in sectors^^
!! !!
!! ^^=00 = file active  other values (e.g. 01) = user #
!!      E5 = file erased
!!
^^=displacement of line in directory sector
```

## SECOND LINE

```
10 6162636465000000000000000000000000 abcde.....
```

```

:
:
:
^---allocation group numbers---^

```

As with DDT or ZSID, the numbers on the left are the hexadecimal representation of the bytes, with the ASCII values on the right. This shows that we have a file 'FILENAME.TYP' and it is stored in allocation groups 61-65 (hex) - in this case the allocation groups have printable ASCII values (abcde). The allocation groups don't have to be consecutive, and almost certainly won't be if you have done some deleting on the disk.

The value 23 at the end of the first line is the number of sectors taken up by the file, in this case 23H or 35 sectors. At 8 sectors per group, you can calculate that only 3 sectors of group 65 are used by the file.

One directory entry can have a maximum of 16 allocation groups (16 groups = 16k). If a file is larger than this, it requires a second (or more) directory entry with the extent byte now being set to 01, 02 etc.

One of DU's options is to 'map' the disk, giving a summary of where all files are stored, by allocation group.

Have you ever wondered why the CP/M Master disk came with a function key set-up for SVMBASIC but this programme was not on the disk? Rumours have it that SVMBASIC was on the distribution disk in the northern hemisphere.

Again delving with DU, mapping the CP/M 2.22 master disk shows that SVMBASIC is in fact on the disk, but has been erased (first byte set to E5)!!! It is shown in the Directory on Track 3, Sector 4. The file can easily be recovered by setting the first byte to 00 for both Directory entries (a second extent is used) and writing the change to disk, or can be recovered using the public domain utility UNERASE.COM by typing UNERASE SVMBASIC.COM. We can only wonder why the file was erased - perhaps it is buggy, although it does seem to work.

Also on the CP/M 2.22 Master disk, but erased, are files called:

MEMTST.COM (Directory Track 3, Sector 6)

SELTST.COM ( Track 3, Sector 7)

CLBUFF.BAS ( Track 3, Sector 7)

TEST .COM ( Track 3, Sector 1)

All these can be recovered by one of the ways described above.

MENTST is a programme called MEMR Rasmussen Memory Test - Version 2.0. It does some sort of check of memory, but hangs up part way through. One thing it does tell you is where the BIOS starts in memory (E600).

SELTST doesn't seem to do anything useful, succeeding only in hanging up the computer. TEST behaves similarly.

CLBUFF gives a basic routine for clearing the type-ahead (i.e. keyboard) buffer - nothing new here.

Delving with DU on the CP/M 2.23 disk is much more interesting. Stepping through the 'free' allocation groups reveals a wealth of information. There is the source code (i.e. ASM files) for SVCOPY.ASM, SVFMT2.ASM and TRANSFER.MAC. If you wanted to know how to go about writing programmes such as these, here is your chance, that is assuming you have CP/M 2.23. You may also be able to pick out useful routines for using in your own programmes.

SVCOPY does a total track copy of a double-sided disk to another double-sided disk; SVFMT2 is the double-sided 40 track formatting programme, and TRANSFER is a programme for transferring files from single-sided to double-sided disks if you have two double-sided drives i.e. a 605B expander.

Finding where the programmes resided on disk took some time, as some of them seem to have been written to disk more than once. Without having assembled and run the programmes, I can only assume that what has been recovered are the final versions.

This is where our newly acquired knowledge of the Directory and file storage becomes useful. We now have to construct a Directory entry for each file. To do this, I am going to assume that you have DU-V83.

Go to Track 3 and step through the Directory sectors until you get to the end of used space - after the last-used rows the sectors will be filled with e5 bytes.

Reconstruct a two row entry for each file as follows. Note that you don't enter the displacement shown on the left - enter the appropriate value to tell DU where to write the information:

|    | <u>Enter this column</u>            | <u>DON'T enter this column</u> |
|----|-------------------------------------|--------------------------------|
| 00 | 00535643 4F505920 2041534D 00000051 | SVCOPY ASM                     |
| 10 | 71727374 7B7C0000 00000000 00000000 |                                |
| 20 | 00545241 4E534645 524D4143 00000060 | TRANSFERMAC                    |
| 30 | 7D7E7F80 81820000 00000000 00000000 |                                |
| 40 | 00535646 4D543220 2041534D 0000008D | SVFMT2 ASM                     |
| 50 | 75767778 797A8586 87000000 00000000 |                                |

After writing this to the disk (use a copy of the Master just in case something goes wrong), the files will be shown by a 'Dir' command, can be typed to screen or printer using the CP/M type command, and can be assembled. You may have to do a bit of editing of TRANSFER to assemble it with ASM.COM because it appears to have been written for a Macro-assembler.

Also on the CP/M 2.23 disk are the following, which are probably of little interest to most of us. A Directory entry can be constructed for them if you want to see what they do. You can give them whatever name you like:

|                  |                                                                            |
|------------------|----------------------------------------------------------------------------|
| MENTST.COM       | Erased file, shown in Directory; T3, S7<br>Recover as for CP/M 2.22        |
| D10MOD.TXT       | Direct CP/M calls from PL/I-80<br>Allocation groups 89-93 hex; A7h sectors |
| MPMDIQ.TXT       | Direct MP/M calls from PL/I-80<br>Allocation groups 94-9B; 73h sectors     |
| DIVBY2           | PL/I-80 routine<br>Group 9C; 0Ah sectors                                   |
| DIVBY2(Function) | PL/I-80 routine<br>Group 9D; 0Dh sectors                                   |
| DSKEDCPM         | CP/M Disk Editor Rev 0.00 of 81-04-01<br>Groups 9E-9F; 19h sectors         |

The remaining 'free' tracks on the disk are essentially junk - bits of over-written files etc.

The disk editor appears to be a very primitive alternative to DU, and I can't get all of it to work. It may be of interest to see how someone has tackled writing such a programme.

The CP/M 2.23 Master disk also has two function keys for SVMBASIC, but this time there's no sign of it on the disk.

If you don't have DU yet, I recommend you get a copy. Not only can you have fun hacking round like this, but also you will find it invaluable for altering system tracks. The following are some of the useful alterations you can make to the system tracks (applies to CP/M 2.22 & 2.23 unless stated otherwise):

# SPECTRAVIDEO 328 -- Alteration of disk system tracks

| <u>Function</u>                                                | <u>Location</u>                                                  |
|----------------------------------------------------------------|------------------------------------------------------------------|
| Boot-up message                                                | T1, S3                                                           |
| Programme to run automatically on cold boot                    | T1, S7, Bytes 8-16                                               |
| Spin patch: CP/M 2.22<br>CP/M 2.23                             | T1, S1, Bytes 39, 3A<br>Bytes 48, 49                             |
|                                                                | To stop spin, bytes are 32, 00<br>No spin stop, bytes are 08, 07 |
| Function keys assignments                                      | T0, S3, S5, S7                                                   |
| Converting drives from single-<br>to double sided -- CP/M 2.23 | T0, S7, Byte 43 (A: drive)<br>Byte 53 (B: drive)                 |

The byte should be 59 if a single-sided drive, or 77 for double-sided.  
This points to the appropriate disk parameter table, at E759 or E777.

DU can also be used to read and write to Basic disks. I have successfully repaired the Directory and File Allocation table in this way. More on this later, if anyone is interested and I can remember what I did.

Harry Patterson from Nelson has defected to the join the Matthews Gang (AMIGA owners) and is selling his SV setup

SV 328 Mk 1 ✓  
Mini Expander  
2 Disk Drives ✓  
2 Centronics Interfaces (1 Spare)  
80 Column Card ✓  
Cassette deck ✓  
Printer cable ✓  
100 S/S Disks} All full of Basic ✓  
70 Cassettes} and CP/M programmes ✓  
2 Cartridges  
All SV books that were available ✓  
All Wellington and Auckland Magazines ✓  
A few MSX books including Programmers Guide

Selling preferably as a complete lot for \$1200

Contact Harry at 52 Britannia Heights, NELSON Tel. 83804

\*\*\*\*\*

And from Collan Cowan, a fairly useful programme

```
10 ' LOADER PROGRAM.
20 ' THIS WILL ELIMINATE THE NEED FOR A LOADER PROGRAM ON
    GAMES SUCH AS ARMoured ASSAULT AND SPECTRON.
30 ' THUS ONLY TAKING UP 4 TRACKS OF THE DISK FOR EACH GAME.
40 ' LINE 120, CHANGE "1:AAss " TO THE DISK NUMBER YOU ARE
    LOADING FROM AND THE NAME YOU HAVE GIVEN TO THE LOADER
    PROGRAM
50 ' LINE 220, CHANGE "1:ARMOUR" TO THE DISK NUMBER ARE
    SAVING TO AND THE NAME YOU WISH TO SAVE THE PROGRAM
    UNDER.
60 '
70 CLEAR 200,&H87FF
80 FOR A=&H8800 TO &H880E
90 READ A$
100 POKE A,VAL("&H",+A$)
110 NEXT
120 LOAD "1:AAss"
130 OUT &H81,0:OUT &H81,0
140 FOR HL=&HC000 TO &HD7FF
150 POKE HL-(&HC000-&H880F),INP(&H84)
160 NEXT
170 OUT &H81,0:OUT &H81,&H20
180 FOR HL=&HD800 TO &HF7FF
190 POKE HL-(&HC000 TO &H880F),INP(&H84)
200 NEXT
210 BEEP:SCREEN 0:PRINT"Press any key to save...";
    :Z$=INPUT$(1)
220 BSAVE "1:ARMOUR",&H8800,&HC000
230 DATA F3,21,0F,88,11,00,C0,01,00,38,ED,B0,C3,00,C0
```

MODEMS MODEMS MODEMS MODEMS MODEMS MODEMS MODEMS MODEMS

A recent Waikato Users Newsletter advertised a modem suitable for the Spectravideo 300 and 700 series which their members have produced. It costs \$120.00 and includes the board, a transformer and full instructions. Other components are available at most electronics outlets. Software is available for the cost of a disk. Further details from the Group:

Waikato Users Group  
P.O.Box 16-113  
Glenview  
Hamilton

JOHN HODGSON has already bought a package from them and is interested in hearing from anyone else in our group who is contemplating buying a modem. (You will need to have an RS 232 card of course!) You can contact John at 70 Gloaming Hill, Titahi Bay.

COLIN CHIN advised the Group some time ago that he was purchasing a modem from the Brisbane Group (Brian Parker) and was going to apply for NZPO approval. It has been submitted and he awaits a response from the Post Office.

We received advice that someone in Otago had produced a modem which was an RS 232/modem combination and sold for about \$150 including the software to run it. Can anyone give further details or have personal experience of this device?

\*\*\*\*\*

Collan Cowan from Wanganui wrote recently requesting the following software:

SD 260T Mini Golf  
SD 916T Golf  
SD 348T Jumping Jack

If YOU can help please contact him:

E.N. Cowan  
11/Pukenamu Flats  
WANGANUI

\*\*\*\*\*

## PRINTER CODES on the P.D. TEXT EDITOR by Steve Lacey

I have had several queries from members saying this program does everything that a normal text editor should do except, provide the means of changing text style using the printer codes. *N O T T R U E!*  
Below is a list of the programs features.

- no 80 column card needed.
- 40 to 255 columns text with screen scroll.
- can't loose text easily.
- cassette based, save and load.
- everything you need to know is on F keys.
- word character search.
- line insert and line delete.
- Printer codes.

The printer codes are inserted into text using the function key F8. First place the cursor where you want a change in the style. I find that if I type in all my text first it is easier to insert the codes later. I set my column width to about 10 columns larger and you will hear a BEEP at this point. This allows me to right justify the lines myself & when the codes are put into the text, it will extend the line further right on the screen, but not on the printout.  
Note you must create space to insert codes.

How to use the codes. *How to use codes. Use the codes.*

This program only accepts decimal codes. This text is printed on a CP80 printer and the codes are on page 36 of the manual. Other printers may not use the same numbers. The easiest CP80 codes to use are)

- 14 - Turns on DOUBLE width mode for 1 LINE (.)
- 15 - Turns on COMPRESSED character Mode (//)
- 18 - Turns off COMPRESSED character Mode (2)

The symbols I have displayed at the end of the above lines are the codes that are inserted into text. You can't make the printouts work unless it is done through the F8 key for each number.

The control codes which use <ESC> means that 27 has to be entered first. A semicolon appears where the cursor was and you proceed with F8 again & enter the next No.

- 27 52 - Turns on *Italic lettering*
- 27 53 - Turns off *Italic lettering*
- 27 83 - Turns on Super script *super script*
- 27 84 - Turns off Super script
- 27 64 - Resets all modes (use after mixed codes)
- 27 60 - One line print from left to right (alignment)
- 27 69 - Double strike Mode on
- 27 70 - Double strike mode off

Have fun and try mixing codes. *MIXED CODES*

This program is on the Public Domain Tape No 6 along with 14 other programs, most of which are printer based.

G. CLARK

## A CP/M PRINTER SETUP for Epson FX printers

by Graeme Dick

When using a programme such as Wordstar I have often found it a nuisance not to be able to use many of the printer features that are now available. This became even more apparent recently with the purchase of a Citizen 120D printer, which can function as an Epson FX printer or as an IBM Graphics Printer (software selectable) - incidentally, it is a very good buy for your money in my opinion. Many of the dot matrix printers available emulate some model of Epson, and if yours does then some of the commands in the programme will work for you.

There are public domain Wordstar patches which presumably enable extra commands to be embedded in your text. Not having such a patch available, I have written the following programme in 8080 assembly language to enable global commands (that is, commands affecting the entire text) to be set up by calling the programme from within Wordstar using the "R" option. The programme can, of course, also be used as a standard .COM file.

The programme is largely explained by comments included in the assembly file - the Source (ASM) and Object (COM) files are available from the Wellington User Group. Type it in, save it as a .ASM file, and then assemble it in the usual way - e.g. ASM PRINTER.ABC (where the A specifies the drive containing the source file, B the drive that will receive the HEX file, and C the destination of the PRN file - see your CP/M manual if not familiar with these. If you prefer Z80 code and have a Z80 assembler, it is quite easy to convert the ASM file to Z80 code.

The codes used are as follows:

|                    |            |                       |             |
|--------------------|------------|-----------------------|-------------|
| Elite              | = esc "M"  | Line spacing 7/72"    | = esc "1"   |
| Pica               | = esc "P"  | Line spacing 1/6"     | = esc "2"   |
| Compressed ON      | = 0Fh      | Uni-directional print | = esc "U1"  |
| Compressed OFF     | = 12h      | Bi-directional print  | = esc "U0"  |
| Proportional ON    | = esc "p1" | Slashed zero ON       | = esc "~41" |
| Proportional OFF   | = esc "p0" | Slashed zero OFF      | = esc "~40" |
| Correspondence ON  | = esc "x1" | Paper-out sensor ON   | = esc "9"   |
| Correspondence OFF | = esc "x0" | Paper-out sensor OFF  | = esc "8"   |
| RESET              | = esc "@"  |                       |             |

This is the display that you will see when running 'Printer', with the default settings indicated:

Spectravideo CP/M 2.2x EPSON FX Printer Initialisation  
 Donated to the Public Domain by G L Dick New Zealand 25/9/86

# **INITIALISATION OF GLOBAL COMMANDS**

|                            |              |                |                       |     |
|----------------------------|--------------|----------------|-----------------------|-----|
| Elite                      | 12 cpi       | A              | Line spacing 7/72"    | I   |
| Pica                       | 10 cpi       | B <            | Line Spacing 1/6"     | J < |
| Compressed ON              | 20 or 17 cpi | C              | Uni-directional print | K   |
| Compressed OFF             |              | D <            | Bi-directional print  | L < |
| Proportional ON            |              | E              | Slashed zero ON       | M   |
| Proportional OFF           |              | F <            | Slashed zero OFF      | N < |
| Correspondence quality ON  |              | G              | Paper-out sensor ON   | O < |
| Correspondence quality OFF |              | H <            | Paper-out sensor OFF  | P   |
| RESET TO DEFAULT SETTINGS  | ESC          | Finished ..... |                       | ^C  |

## **Enter Your choice**

The Printer programme gives an easy introduction to assembly programming, using essentially just three of CP/M's built-in functions:

- Function 1: Console Input, which reads a keyboard character and stores it in register A. This functions by loading Reg C with 01h then calling the BDOS (entry point=0005h).
- Function 5: List Output, which sends a character to the LST device, which should be the printer. This functions by loading Reg C with 05h, Reg E with the character to be sent, and then calling BDOS.
- Function 9: Print String, which prints a string to the console device. Reg C is loaded with 09h and Regs DE with the address of the string. Strings are terminated by a '\$', and various formatting commands can be embedded in the string. You can, for example, include 'esc "p"' and 'esc "q"' to turn reverse video on and off, and 'esc "Y", row, column, ' to locate the cursor. Note that the row and column addresses for cursor location have a 20h offset - the top left of the screen is 20h,20h.

Incidentally, after assembly you can alter string characters using DDT or ZSID so that graphics characters are printed to the screen. I have, for example, altered the '<' character to D5h, the back arrow. The 8080 assembler will not recognise the graphics characters.

P L E A S E  
R E M E M B E R  
Y O U R  
S U B S C R I P T I O N

A plea from your committee!

There are a number of outstanding subscriptions dating back over two months in some cases.

As it is your subscriptions which keep the club running, if you want the support of the Group to continue, please help us by checking your subscription renewal date.

Your subscription renewal date is printed on the address label of every newsletter.

As the Group does not send out any other reminders for subscriptions, we have been allowing some latitude with issues being produced every two months.

If your subs are due before say January, please forward them to the Club's P.O. Box 26050 - Newlands, WELLINGTON.

Remember - the subs are only:

\$18.00 For NZ Residents  
NZ\$30 For Australasian Subscriptions  
NZ\$40 For other areas overseas.

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Another Plea . . . . .

R A D I O   A C C E S S  
BROADCASTS FOR  
C H R I S T M A S

As its getting near THAT time of the year again, we want to make our last 'Broadcast' for 1986 appropriate to the season.

If anyone has a lack of Inspiration how about some

Christmas Music  
Christmas Scenes (Drawing program screen saves or whatever)  
Holiday-time Programs

for the December Radio Access Broadcast. Please contact Garry, or send something along to our November meeting.